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VALVE SEALING STRUCTURES

Abstract

Valve sealing structures are disclosed. An example apparatus includes a valve body including a seat, and a valve plug including a disc holder including a recess; a seat ring positioned in the recess, the seat ring including a first surface and a second surface contiguous with the first surface, the first surface to contact the seat when the valve plug is in a closed position; and a disc cover coupled to the disc holder, the disc cover including a third surface and a fourth surface contiguous with the third surface, the third surface to be flush with the first surface, the fourth surface to contact the second surface and apply an axial pre-load force to the seat ring.

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Background/Summary

RELATED APPLICATIONS

[0001] This patent claims the benefit of Chinese Patent Application No. 202420453732.5, which was filed on Mar. 8, 2024, and Chinese Patent Application No. 202411172861.8, which was filed on Aug. 23, 2024. Chinese Patent Application No. 202420453732.5 and Chinese Patent Application No. 202411172861.8 are hereby incorporated herein by reference in its entirety. Priority to Chinese Patent Application No. 202420453732.5 and Chinese Patent Application No. 202411172861.8 is hereby claimed.

FIELD OF THE DISCLOSURE

[0002] This disclosure relates generally to valves and, more particularly, to valve sealing structures.

BACKGROUND

[0003] In recent years, disc seals (e.g., disc-shaped sealing structures, seat rings, etc.) have been used in valves to control fluid flow through an opening. The disc seal can be part of a plug that blocks a middle portion of the opening, and the disc seal can define a perimeter of a portion of the plug and contact a seat (e.g., a plug seat) to block a perimeter of the opening. Accordingly, the disc seal prevents fluid from flowing through the opening when the valve is in the closed position.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. 1 illustrates an example valve including a disc holder, a seat ring, and a disc cover in accordance with examples disclosed herein.

[0005] FIG. 2 illustrates the valve of FIG. 1 in a first example position.

[0006] FIG. 3 illustrates the valve of FIG. 1 in a second example position.

[0007] FIG. 4 illustrates a magnified view of a portion of the valve of FIGS. 1-3 in the second example position of FIG. 3.

[0008] FIG. 5 illustrates an isolated view of the seat ring of FIGS. 1-4.

[0009] FIG. 6 illustrates an isolated view of the disc cover of the valve of FIGS. 1-3.

[0010] FIG. 7 illustrates an isolated view of the disc holder of the valve of FIGS. 1-3.

[0011] FIG. 8 illustrates another example seat ring that can be implemented in the valve of FIG. 1 in accordance with examples disclosed herein.

[0012] FIG. 9 illustrates another example seat ring that can be implemented in the valve of FIG. 1 in accordance with examples disclosed herein.

[0013] FIG. 10 illustrates another example seat ring that can be implemented in the valve of FIG. 1 in accordance with examples disclosed herein.

[0014] FIG. 11 illustrates another example seat ring that can be implemented in the valve of FIG. 1 in accordance with examples disclosed herein.

[0015] FIG. 12 illustrates another example seat ring that can be implemented in the valve of FIG. 1 in accordance with examples disclosed herein.

[0016] In general, the same reference numbers will be used throughout the drawing(s) and accompanying written description to refer to the same or like parts. The figures are not necessarily to scale.

DETAILED DESCRIPTION

[0017] A valve typically includes a seat ring (e.g., a seal, a disc sealing structure) to contact a seat of a valve body (e.g., a perimeter of a conduit). The seat ring prevents fluid from flowing between the seat and the seat ring when the valve is in a closed position. For example, a plug of the valve can include the seat ring, a disc holder, and a disc cover. The disc holder can include a slot (e.g., a

recess) for the seat ring. Further, the disc cover can couple to the disc holder and compress the seat ring to provide an axial pre-load force to the seat ring. Pre-loading the seat ring improves the sealing capabilities associated therewith. For example, pre-loading can cause the seat ring to conform to irregularities and imperfections in contacted surfaces to ensure a tighter seal and reduce leakage. Additionally, pre-loading can help reduce seal or gasket relaxation, improve load distribution and resilience to temperature changes, and prevent movement of the seat ring to increase a pressure at which the seat ring can maintain sealing capabilities.

[0018] During operation, the valve plug can move (e.g., translate) along an axis (e.g., a longitudinal axis of the valve) to move the seat ring toward and away from the seat to control a flow of fluid past the valve. In some instances, a surface of the disc cover is positioned at an angle relative to the longitudinal axis of the valve such that the surface at least partially faces the conduit through which fluid flow is blocked when the valve is in the closed position.

[0019] In some known implementations, a surface of the seat ring that contacts the seat when the valve is in the closed position (e.g., a sealing surface) is positioned at a same angular orientation as the first surface of the disc cover. However, in some such implementations, adjacent ends of the sealing surface and the surface of the disc cover are separated by a gap such that a boundary of the flow path defined by seat ring and the disc cover is not level, which causes the fluid flowing through the valve to exhibit undesirable flow properties, such as a vortex or other inefficient and/or unpredictable flow.

[0020] In some other known implementations, the sealing surface abuts the surface of the disc cover. However, in such known implementations, another surface (e.g., a non-sealing surface) of the seat ring contiguous with the sealing surface is substantially parallel to the longitudinal axis of the valve to enable the sealing surface to abut the surface of the disc cover. As a result, a relatively small angle is defined at a tip of the seat ring between the sealing surface and the non-sealing surface. Several factors can cause the tip to rupture or protrude from the slot of the disc holder, which negatively affects sealing capabilities and/or fluid flow. For instance, the rupture or protrusion can result from a difference in area between the seat ring and a portion of the disc cover that extends through the seat ring. Additionally or alternatively, the rupture or protrusion can result from thermal expansion of valve components, such as an O-ring that contacts the seat ring.

[0021] Examples disclosed herein provide valve plugs that define smooth flow path boundaries while also maintaining sealing capabilities at relatively higher pressures. Examples disclosed herein include seat rings that have a sealing surface aligned along a same geometric plane as and abutting (e.g., flush with) an adjacent surface of the disc cover. The example seat rings disclosed herein also have a non-sealing surface that is contiguous with the sealing surface and that is nonparallel to the longitudinal axis of the valve. As such, the non-sealing surface receives a pre-load force from the disc cover proximate an end of the sealing surface, which increases a contact area across which the disc cover applies the pre-load to the seat ring. Thus, the seat ring receives the pre-load across a greater portion of a width of the seat ring (e.g., between an inner diameter and outer diameter thereof). As a result, the seat ring can provide sealing at relatively higher pressures while also providing a smooth flow path boundary at a transition between the seat ring and the disc cover to minimize undesirable flow properties.

[0022] Turning to the figures, FIG. 1 illustrates an example valve **100** including a valve body **102** and a valve plug **104** (e.g., a seat sealing structure). The valve plug **104** includes a valve stem **105**, a disc holder **106**, a seat ring **108** (e.g., a seal or gasket), and a disc cover **110**. The valve body **102** includes an inner wall **112** and a seat **114** (e.g., a plug seat) positioned around an orifice. In some examples, the seat **114** is positioned at an end of the inner wall **112**.

[0023] In the illustrated example of FIG. 1, the seat ring **108** is positioned between the disc holder **106** and the disc cover **110**. Specifically, the disc holder **106** includes a recess **116** (e.g., a recessed portion, a slot, etc.) in which the seat ring **108** is positioned. The seat ring **108** is disc-shaped (e.g., ring-shaped). A first portion **118** (e.g., an inner portion, a lower portion in the orientation of the

valve **100** in FIG. 1) of the disc cover **110** extends through an inner diameter (e.g., an opening) of the seat ring **108** and couples to the disc holder **106**. In this example, the first portion **118** includes threads **119** to couple the disc cover **110** to the disc holder **106**.

[0024] In the illustrated example of FIG. 1, a second portion **120** (e.g., an outer portion, an upper portion in the orientation of the valve **100** in FIG. 1) contacts a surface (e.g., an upper surface) of the seat ring **108** and applies an axial pre-load force to the seat ring **108**, as discussed in further detail below. As a result, the seat ring **108** is compressed between the disc holder **106** and the disc cover **110** to enhance sealing capabilities associated therewith. For example, a rotation of the disc cover **110** to threadedly couple the disc cover **110** to the disc holder **106** can result in linear movement of the disc cover **110** towards the disc holder **106** and the seat ring **108** to compress the seat ring **108** against the disc holder **106**.

[0025] In the illustrated example of FIG. 1, the valve plug **104** is movable between (i) a first position (e.g., an open position) in which the seat ring **108** is separated from the seat **114** to enable fluid to flow therebetween and (ii) a second position (e.g., a closed position) in which the seat ring **108** contacts the seat **114** to block or prevent a fluid from flowing through the orifice in the valve body **102**. Specifically, when the valve plug **104** is in the first position, the valve plug **104** opens a flow path between the seat ring **108** and the seat **114** to enable fluid to flow from the first conduit **122** to the second conduit **124**, as discussed further in association with FIG. 2. When the valve plug **104** is in the second position, the valve plug **104** blocks fluid in a first conduit **122** of the valve body **102** from flowing past the seat **114** and into a second conduit **124** downstream of the first conduit **122**, as discussed further in association with FIGS. 3 and 4. In some examples, the position of the valve plug **104** can be modulated to one or more intermediate positions between the first position and the second position to adjust a space provided between the seat ring **108** and the seat **114** and, in turn, adjust a flow rate of fluid flowing from the first conduit **122** to the second conduit **124**.

[0026] In the illustrated example of FIG. 1, the valve **100** includes a spring **126** to facilitate movement of the valve plug **104**. Specifically, the spring **126** urges the valve plug **104** along an axis **128** (e.g., a longitudinal axis of the valve plug **104**) between the first position and the second position. That is, the valve plug **104** is translatably coupled to the valve body **102**. In some examples, the spring **126** moves the valve plug **104** to the second position when a pressure downstream of the valve plug **104** satisfies (e.g., is greater than, is greater than or equal to) a pressure threshold. For example, when the pressure satisfies the pressure threshold, the spring **126** can be released from a biased configuration to translate the valve plug **104** and press the seat ring **108** against the seat **114** to close a flow path defined by the valve **100**. In some other examples, the valve **100** includes another actuator (e.g., a servo motor) to translate the valve plug **104** along the axis **128** between the first position and the second position based on a flow rate to be obtained.

[0027] FIG. 2 illustrates the valve **100** of FIG. 1 including the valve plug **104** in an example open position **200**. In FIG. 2, the seat ring **108** is separated from the seat **114** to enable fluid to flow therebetween. In FIG. 2, an orifice **202** defined by the seat **114** is aligned along a geometric plane **204** that is substantially perpendicular to the axis **128** along which the valve plug **104** translates. In this example, the orifice **202** has a diameter of approximately 10 millimeters (mm.).

[0028] FIG. 3 illustrates the valve **100** of FIG. 1 including the valve plug **104** in an example closed position **300**. In the closed position **300** of FIG. 3, the seat ring **108** contacts the seat **114** to block fluid from flowing therebetween (e.g., through the orifice **202** (FIG. 2)). For example, the valve plug **104** can translate along the axis **128** to move from the open position **200** of FIG. 2 to the closed position **300** of FIG. 3 to reduce the flow rate of the fluid flowing from the first conduit **122** to the second conduit **124**.

[0029] FIG. 4 illustrates a magnified view of a portion of the valve **100** of FIGS. 1-3 including the valve plug **104** in the example closed position **300** of FIG. 3. Accordingly, the seat ring **108** is in contact with the seat **114**. In some examples, when the valve plug **104** is in the closed position **300**,

the seat ring **108** is compressed against the seat **114** to increase a contact area therebetween and maintain the seal at relatively higher pressures. In some such examples, the seat ring **108** encounters elastic deformation.

[0030] In the illustrated example of FIG. **4**, the seat ring **108** includes a first surface **402** (e.g., a first face, a first bevel), a second surface **404** (e.g., a second face, a second bevel), a third surface **406** (e.g., a third face), a fourth surface **408** (e.g., a fourth face), a fifth surface **410** (e.g., a fifth surface), and a sixth surface **412**. The first surface **402** contacts the seat **114** when the valve plug **104** is in the closed position **300**. The second surface **404**, the third surface **406**, and the fourth surface **408** are in contact with the disc cover **110**. The fifth surface **410** and the sixth surface **412** are in contact with the disc holder **106**.

[0031] In the illustrated example of FIG. **4**, the first surface **402** is contiguous with the second surface **404**. Contiguous surfaces herein can be connected by a sharp edge or a curved edge (e.g., a chamfer or fillet) at respective ends of the surfaces. As used herein, a first surface is distinct, and referred to separately, from a second surface that is contiguous with the first surface when there is a change in angular orientation between the first surface and the second surface.

[0032] In the illustrated example of FIG. **4**, the first surface **402** and the second surface **404** include bevels that are oblique relative to the axis **128** along which the valve plug **104** translates. As such, the first surface **402** and the second surface **404** are non-perpendicular and nonparallel to the axis **128**. Similarly, the first surface **402** and the second surface **404** are oblique to the geometric plane **204** (FIG. **2**) defined by the orifice (**202**) around which the seat **114** is positioned. In some examples, the first surface **402** is positioned (e.g., oriented) at a first angle relative to the axis **128** between 15 degrees) ($^{\circ}$ and 90°). In some examples, the second surface **404** is positioned (e.g., oriented) at a second angle relative to the axis **128** between 0° and 150° . For example, the first angle can be approximately 45° relative to the axis **128** in a first direction, and the second angle can be approximately 45° relative to the axis **128** in a second direction opposite the first direction.

[0033] In some examples, the third surface **406** is contiguous with the second surface **404**; the fourth surface **408** is contiguous with the third surface **406**; the fifth surface **410** is contiguous with the fourth surface **408**; and/or the sixth surface **412** is contiguous with the first surface **402** and/or the fifth surface **410**. In some examples, the third surface **406** and/or the fifth surface **410** are substantially parallel to the axis **128** along which the valve plug **104** translates. In some examples, at least a portion of the fourth surface **408** is substantially perpendicular to the axis **128**.

[0034] In the illustrated example of FIG. **4**, the disc cover **110** includes a seventh surface **414** (e.g., a seventh face, a third bevel surface), an eighth surface **416** (e.g., an eighth face, a fourth bevel surface), a ninth surface **418** (e.g., a ninth face), and a tenth surface **420**. In some examples, the seventh surface **414** is contiguous with the eighth surface **416**; the eighth surface **416** is contiguous with the ninth surface **418**; and/or the ninth surface **418** is contiguous with the tenth surface **420**. The seventh surface **414** is flush with the first surface **402**. That is, the first surface **402** and the seventh surface **414** abut and are aligned along a same geometric plane **424**. Accordingly, the seventh surface **414** is positioned (e.g., oriented) at the first angle relative to the axis **128** (e.g., a same angle as the first surface **402**). As a result, a transition area **422** in a flow path boundary between the seat ring **108** and the disc cover **110** (e.g., between the first surface **402** and the seventh surface **414**) does not generate a vortex in the fluid flow or otherwise affect flow characteristics in an undesirable manner (e.g., an unpredictable manner, a manner that reduces controllability of the flow properties) as the fluid flows across the transition area **422**.

[0035] In the illustrated example of FIG. **4**, the eighth surface **416** is in contact with the second surface **404**, and the ninth surface **418** is in contact with the third surface **406**. The eighth surface **416** is positioned at the second angle relative to the axis **128** (e.g., a same angle as the second surface **404**). In some examples, the ninth surface **418** is positioned substantially perpendicular to the axis **128**. In such examples, an angle of the eighth surface **416** relative to the ninth surface **418** is greater than 90° . In some examples, the tenth surface **420** is substantially parallel to the axis **128**.

[0036] In the illustrated example of FIG. 4, the eighth surface **416** and the ninth surface **418** apply an axial pre-load force to the seat ring **108**. That is, the eighth surface **416** and the ninth surface **418** apply a force that compresses the seat ring **108** against the disc holder **106**. Specifically, the eighth surface **416** applies a first force to the second surface **404** that is normal to the second surface **404**, and the ninth surface **418** applies a second force to the third surface **406** that is normal to the third surface **406**. Accordingly, the first force is oblique to the axis **128**, and the second force is substantially parallel to the axis **128**. As such, the threads **119** (FIG. 1) of the disc cover **110** enable conversion of a rotation of the disc cover **110** to linear movement that applies the first force and the second force.

[0037] Advantageously, the angular orientation of the second surface **404** and the eighth surface **416** increases a surface area across which the disc cover **110** can apply the pre-load to the seat ring **108**. Further, the surfaces **404**, **416** contiguous with the first surface **402** and the seventh surface **414**, respectively, can receive and apply the pre-load proximate the first surface **402** while the first surface **402** is flush with the seventh surface **414**. As such, the seat ring **108** and the disc cover **110** enable an increased pre-load to be applied to the seat ring **108** for improved sealing of the orifice **202** (FIG. 2) (e.g., sealing at relatively higher pressures) while avoiding a negative effect on the fluid flow properties.

[0038] In the illustrated example of FIG. 4, the disc holder **106** includes an eleventh surface **426** (e.g., an eleventh face), a twelfth surface **428** (e.g., a twelfth face), and a thirteenth surface **430** (e.g., a thirteenth face). In some examples, the eleventh surface **426** is contiguous with the twelfth surface **428** and the thirteenth surface **430**. In some examples, the eleventh surface **426** is substantially parallel to the axis **128**. In some examples, at least a portion of the twelfth surface **428** is substantially perpendicular to the axis **128**. The thirteenth surface **430** defines a portion of a boundary of the flow path in the valve body **102**. The eleventh surface **426** and the twelfth surface **428** define a recess **432** (e.g., a slot or groove) in the disc holder **106** in which the seat ring **108** is positioned. Specifically, the eleventh surface **426** is in contact with the sixth surface **412**, and the twelfth surface **428** is in contact with the fifth surface **410**.

[0039] In the illustrated example of FIG. 4, the fifth surface **410** includes raised portions **434** (e.g., raised edges, recesses, indents), and the twelfth surface **428** includes protrusions **436** that extend past a portion (e.g., a planar portion) of the fifth surface **410** and into cavities defined by the raised portion **434**. Specifically, the raised portions **434** can define a first serrated portion of the fifth surface **410**, and the protrusions **436** can define a second serrated portion of the twelfth surface **428** that meshes with the first serrated portion. For example, the raised portions **434** and the protrusions **436** can define concentric serrations. Advantageously, the raised portions **434** and the protrusions **436** block leakage of the fluid between the fifth surface and the twelfth surface **428**, which can remove a need for an O-ring between a bottom surface (e.g., the fifth surface **410**) of the seat ring **108** and the disc holder **106**. Moreover, an O-ring typically requires a seat ring (e.g., the seat ring **108**) to have a relatively longer width (e.g., between the fourth surface **408** and the sixth surface **412**) to provide enough contact area between the seat ring and the O-ring to prevent such leakage. As such, the raised portions **434** and the protrusions **436** also enable a size of the seat ring **108**, and, in turn, the valve plug **104**, to be reduced. Although the illustrated example of FIG. 4 depicts two of the raised portions **434** and corresponding protrusions **436** positioned therein, the fifth surface **410** and the twelfth surface **428** can include a different number of raised portions **434** and protrusions **436**, respectively. For example, the fifth surface **410** and the twelfth surface **428** can include one raised portion **434** and one protrusion **436**, respectively. Alternatively, the fifth surface **410** and the twelfth surface **428** can include more than two raised portions **434** and more than two protrusions **436** positioned therein.

[0040] In some examples, the seat ring **108** includes a first temporary face **438** (e.g., a first pre-installation face) between the third surface **406** and the fourth surface **408**; a second temporary face **440** (e.g., a second pre-installation face) between the fourth surface **408** and the fifth surface **410**;

and a third temporary face **442** (e.g., a third pre-installation face) between the fifth surface **410** and the sixth surface **412**. In such examples, the temporary faces **438**, **440**, **442** flex when the disc cover **110** compresses the seat ring **108** against the disc holder to fill gaps between (i) the seat ring **108** and the disc holder **106** and (ii) between the seat ring **108** and the disc cover **110**. Accordingly, when the temporary faces **438**, **440**, **442** flex, the temporary faces **438**, **440**, **442** change to (e.g., deform, morph into) a curved edge between (i) the third surface **406** and the fourth surface **408**, (ii) the fourth surface **408** and the fifth surface **410**, and (iii) the fifth surface **410** and the sixth surface **412**. As a result, the third surface **406** becomes contiguous with the fourth surface **408**; the fourth surface **408** becomes contiguous with the fifth surface **410**; and the fifth surface **410** becomes contiguous with the sixth surface **412** when the disc cover **110** compresses the seat ring **108** against the disc holder **106**.

[0041] FIG. 5 illustrates an isolated view of the seat ring **108** including the first surface **402**, the second surface **404**, the third surface **406**, the fourth surface **408**, the sixth surface **412**, the first temporary face **438**, the second temporary face **440**, and the third temporary face **442**. FIG. 6 illustrates an isolated view of the disc cover **110** including the seventh surface **414**, the eighth surface **416**, the ninth surface **418**, and the tenth surface **420**. FIG. 7 illustrates an isolated view of the disc holder **106** including the eleventh surface **426**, the twelfth surface **428**, the thirteenth surface **430**, the recess **432**, and the protrusions **436**.

[0042] FIG. 8 illustrates another example valve **800** including another example valve body **802** and another example valve plug **804**. The valve plug **804** includes a disc holder **806**, a seat ring **808**, and a disc cover **810**. The valve body **802** includes an inner wall **812** and a seat **814** positioned around an orifice **816**. The valve **800** is substantially similar to the valve **100** of FIGS. 1-4 except that the orifice **816** is larger than the orifice **202** (FIG. 2) of the valve **100** of FIGS. 1-4. As a result, the disc holder **806**, the seat ring **808**, and the disc cover **810** include an increased width (e.g., a larger diameter) relative to the valve plug **104** of FIGS. 1-4 (e.g., relative to the disc holder **106** (FIGS. 1-4 and 7), the seat ring **108** (FIGS. 1-5), and the disc cover **110** (FIGS. 1-4 and 6)) to block the orifice **816** when the valve plug **804** is in a closed position (e.g., the closed position **300** of FIGS. 3-4). As a result of the increased width, the valve plug **804** of FIG. 8 includes an O-ring **818** positioned between a portion of a first surface **820** (e.g., a first face, a lower surface) of the seat ring **808** and a second surface **822** (e.g., a second face, an upper surface) of the disc holder **806**, in addition to or instead of the raised portions **434** (FIG. 4) and the protrusions **436** (FIGS. 4 and 7), to block fluid from flowing between the first surface **820** and the second surface **822**.

[0043] FIG. 9 illustrates an isolated view of another example seat ring **902** and disc cover **904** of a valve plug that can be implemented in the valve body **102** of FIG. 1 and/or the valve body **802** of FIG. 8 in accordance with examples disclosed herein. In the illustrated example of FIG. 9, the seat ring **902** includes a first surface **906** (e.g., a first face), a second surface **908** (e.g., a second face), a third surface **910** (e.g., a third face), a fourth surface **912** (e.g., a fourth face), a fifth surface **914** (e.g., a fifth face), a sixth surface **916** (e.g., a sixth face), and a seventh surface **918** (e.g., a seventh face). In the illustrated example of FIG. 9, the first surface **906** is contiguous with the second surface **908** and the seventh surface **918**. In some examples, the second surface **908** is contiguous with the third surface **910**; the third surface **910** is contiguous with the fourth surface **912**; the fourth surface **912** is contiguous with the fifth surface **914**; the fifth surface **914** is contiguous with the sixth surface **916**; and/or the sixth surface **916** is contiguous with the seventh surface **918**.

[0044] In the illustrated example of FIG. 9, the first surface **906** is oblique relative to the axis **128** along which the seat ring **902** and the disc cover **904** translate. The second surface **908**, the fourth surface **912**, and/or the sixth surface **916** are substantially perpendicular to the axis **128**. The third surface **910**, the fifth surface **914**, and/or the seventh surface **918** are substantially parallel to the axis **128**.

[0045] In the illustrated example of FIG. 9, the first surface **906** contacts a seat (not shown) of a valve body (e.g., the seat **114** of FIGS. 1-4, the seat **814** of FIG. 8) when the valve plug associated

with the seat ring **902** is in a closed position (e.g., the closed position **300** of FIGS. 3-4). Accordingly, fluid contacts and flows across the first surface **906** when the valve plug is in the open position (e.g., the open position **200** of FIG. 2). The sixth surface **916** and the seventh surface **918** contact a disc holder of the valve plug (e.g., the disc holder **106** of FIGS. 1-4 and 7). In some examples, the valve plug includes an O-ring positioned between a portion of the sixth surface **916** and the disc holder. In some examples, the sixth surface **916** and the disc holder include serrations (e.g., the raised portions **434** (FIG. 4) and the protrusions **436** (FIGS. 4 and 7)), as discussed above. The second surface **908**, the third surface **910**, the fourth surface **912**, and the fifth surface **914** are in contact with the disc cover **904**, as discussed in further detail below.

[0046] In the illustrated example of FIG. 9, the disc cover **904** includes an eighth surface **920** (e.g., an eighth face), a ninth surface **922** (e.g., a ninth face), a tenth surface **924** (e.g., a tenth face), an eleventh surface **926** (e.g., an eleventh face), and a twelfth surface **928** (e.g., a twelfth face). In FIG. 9, the ninth surface **922** is contiguous with the eighth surface **920** and the tenth surface **924**, and the eleventh surface **926** is contiguous with the tenth surface **924** and the twelfth surface **928**.

[0047] In the illustrated example of FIG. 9, the eighth surface **920** is oblique relative to the axis **128**. Specifically, the eighth surface **920** is flush with (e.g., abuts and is positioned along a same geometric plane (e.g., the geometric plane **424** (FIG. 4) as) the first surface **906**. The ninth surface **922** and the eleventh surface **926** are substantially perpendicular to the axis **128**. The tenth surface **924** and the twelfth surface **928** are substantially parallel to the axis **128**.

[0048] In the illustrated example of FIG. 9, the ninth surface **922** is in contact with the second surface **908**; the tenth surface **924** is in contact with the third surface **910**; the eleventh surface **926** is in contact with the fourth surface **912**; and the twelfth surface **928** is in contact with the fifth surface **914**. The ninth surface **922** applies a first pre-load compression force to the second surface **908**, and the eleventh surface **926** applies a second pre-load compression force to the fourth surface **912**. Thus, the disc cover **904** applies the first pre-load compression force to a surface (e.g., the second surface **908**) that is contiguous with a surface (e.g., the first surface **906**) that contacts the seat (e.g., the seat **114** (FIGS. 1-4), the seat **814** (FIG. 8)) to increase a contact area across which the pre-load is applied. As a result, the seat ring **902** receives the pre-load across an increased distance between an inner and outer diameter thereof, which increases the pre-load force received and increases a pressure that the seat ring **902** can withstand during operation. As the second surface **908** is raised relative to the fourth surface **912**, the first and second pre-load compression forces are applied to the seat ring **902** at different heights (e.g., distances from the sixth surface **916**). The first and second pre-load compression forces are substantially parallel to the axis **128** along which the seat ring **902** and the disc cover **904** translate during operation.

[0049] FIG. 10 illustrates an isolated view of another example seat ring **1002** and disc cover **1004** of a valve plug that can be implemented in the valve body **102** of FIG. 1 and/or the valve body **802** of FIG. 8 in accordance with examples disclosed herein. In the illustrated example of FIG. 10, the seat ring **1002** includes a first surface **1006** (e.g., a first face, a first bevel surface), a second surface **1008** (e.g., a second face, a second bevel surface), a third surface **1010** (e.g., a third face), a fourth surface **1012** (e.g., a fourth face), a fifth surface **1014** (e.g., a fifth face), a sixth surface **1016** (e.g., a sixth face), a seventh surface **1018** (e.g., a seventh face), and an eighth surface **1020** (e.g., an eighth face). In the illustrated example of FIG. 10, the first surface **1006** is contiguous with the second surface **1008** and the eighth surface **1020**, and the second surface **1008** is contiguous with the third surface **1010**. In some examples, the third surface **1010** is contiguous with the fourth surface **1012**; the fourth surface **1012** is contiguous with the fifth surface **1014**; the fifth surface **1014** is contiguous with the sixth surface **1016**; the sixth surface **1016** is contiguous with the seventh surface **1018**; and/or the seventh surface **1018** is contiguous with the eighth surface **1020**.

[0050] In the illustrated example of FIG. 10, the first surface **1006** and the second surface **1008** are oblique relative to the axis **128** along which the seat ring **1002** and the disc cover **1004** translate during operation. The third surface **1010**, the fifth surface **1014**, and the seventh surface **1018** are

substantially perpendicular to the axis **128**. The fourth surface **1012**, the sixth surface **1016**, and the eighth surface **1020** are substantially parallel to the axis **128**.

[0051] In the illustrated example of FIG. **10**, the first surface **1006** contacts a seat (not shown) of a valve body (e.g., the seat **114** of FIGS. **1-4**, the seat **814** of FIG. **8**) when the valve plug associated with the seat ring **1002** is in a closed position (e.g., the closed position **300** of FIGS. **3-4**).

Accordingly, fluid contacts and flows across the first surface **1006** when the valve plug is in an open position (e.g., the open position **200** of FIG. **2**). The seventh surface **1018** and the eighth surface **1020** contact a disc holder of the valve plug (e.g., the disc holder **106** of FIGS. **1-4** and **7**). In some examples, the valve plug includes an O-ring positioned between a portion of the seventh surface **1018** and the disc holder. In some examples, the seventh surface **1018** and the disc holder include serrations (e.g., the raised portions **434** (FIG. **4**) and the protrusions **436** (FIGS. **4** and **7**)), as discussed above. The second surface **1008**, the third surface **1010**, the fourth surface **1012**, the fifth surface **1014**, and the sixth surface **1016** are in contact with the disc cover **1004**, as discussed in further detail below.

[0052] In the illustrated example of FIG. **10**, the disc cover **1004** includes a ninth surface **1022** (e.g., a ninth face), a tenth surface **1024** (e.g., a tenth face), an eleventh surface **1026** (e.g., an eleventh face), a twelfth surface **1028** (e.g., a twelfth face), a thirteenth surface **1030** (e.g., a thirteenth face), and a fourteenth surface **1032** (e.g., a fourteenth face). The tenth surface **1024** is contiguous with the ninth surface **1022** and the eleventh surface **1026**. The eleventh surface **1026** is contiguous with the twelfth surface **1028**. The thirteenth surface **1030** is contiguous with the twelfth surface **1028** and the fourteenth surface **1032**. In the illustrated example of FIG. **10**, the ninth surface **1022** and the tenth surface **1024** are oblique relative to the axis **128**. The ninth surface **1022** is flush with (e.g., abuts and is positioned along a same geometric plane (e.g., the geometric plane **424** (FIG. **4**) as) the first surface **1006**. The eleventh surface **1026** and the thirteenth surface **1030** are substantially perpendicular to the axis **128**. The twelfth surface **1028** and the fourteenth surface **1032** are substantially parallel to the axis **128**.

[0053] In the illustrated example of FIG. **10**, tenth surface **1024** is in contact with the second surface **1008**; the eleventh surface **1026** is in contact with the third surface **1010**; the twelfth surface **1028** is in contact with the fourth surface **1012**; the thirteenth surface **1030** is in contact with the fifth surface **1014**; and the fourteenth surface **1032** is in contact with the sixth surface **1016**. The tenth surface **1024** applies a first pre-load compression force to the second surface **1008** in a direction normal to the second surface **1008** (e.g., in a direction oblique relative to the axis **128**). Thus, the disc cover **1004** applies the first pre-load compression force to a surface (e.g., the second surface **1008**) that is contiguous with a surface (e.g., the first surface **1006**) that contacts the seat (e.g., the seat **114** (FIGS. **1-4**), the seat **814** (FIG. **8**)) to increase a contact area across which the pre-load is applied. Further, the eleventh surface **1026** applies a second pre-load compression force to the third surface **1010** in a direction substantially parallel to the axis **128**, and the thirteenth surface **1030** applies a third pre-load compression force to the fifth surface **1014** in the direction substantially parallel to the axis **128**. As a result, the seat ring **1002** receives the pre-load across an increased distance between an inner and outer diameter thereof, which increases the pre-load force received and increases a pressure that the seat ring **1002** can withstand during operation.

[0054] FIG. **11** illustrates a portion of another example valve **1100** including a seat **1102** (e.g., the seat **114** (FIGS. **1-4**), the seat **814** (FIG. **8**)) and a valve plug **1104** in a closed position **1106**. The valve plug **1104** includes another example seat ring **1108**, and another example disc cover **1110**. The seat ring **1108** includes a first surface **1112** (e.g., a first face, a first bevel surface, the first surface **402** of FIG. **4**), a second surface **1114** (e.g., a second face, a second bevel surface), a third surface **1116** (e.g., a third face), a fourth surface **1118** (e.g., a fourth face, the fifth surface **410** of FIG. **4**), and a fifth surface **1120** (e.g., a fifth face, the sixth surface **412** of FIG. **4**). In the illustrated example of FIG. **11**, the first surface **1112** is contiguous with the second surface **1114** and the fifth surface **1120**. In some examples, the third surface **1116** is contiguous with the second surface **1114**

and the fourth surface **1118**. Accordingly, the second surface **1114** spans from an end of the first surface **1112** to an inner diameter of the seat ring **1108**. In some examples, the fourth surface **1118** is contiguous with the fifth surface **1120**. In some examples, the second surface **1114** is contiguous with the fourth surface **1118** at an inner diameter thereof (e.g., the seat ring **1108** does not include the third surface **1116**).

[0055] In the illustrated example of FIG. **11**, the first surface **1112** and the second surface **1114** are oblique relative to the axis **128** along which the valve plug **1104** translates during operation. In some examples, the third surface **1116** and the fifth surface **1120** are substantially parallel to the axis **128**. In some examples, at least a portion of the fourth surface **1118** is substantially perpendicular to the axis **128**.

[0056] In the illustrated example of FIG. **11**, the first surface **1112** contacts the seat **1102** when the valve plug **1104** is in the closed position **1106**. Accordingly, fluid contacts and flows across the first surface **1112** when the valve plug **1104** is in an open position (e.g., the open position **200** of FIG. **2**). The fourth surface **1118** and the fifth surface **1120** are in contact with the disc holder **1106**. In this example, the fourth surface **1118** includes the raised portions **434**, and the disc holder **1106** includes the protrusions **436**, as discussed above. In some other examples, the valve plug **1104** includes an O-ring positioned between at least a portion of the fourth surface **1118** and the disc holder **1106**. The second surface **1114** and the third surface **1116** are in contact with the disc cover **1110**, as discussed in further detail below.

[0057] In the illustrated example of FIG. **11**, the disc cover **1110** includes a seventh surface **1124** (e.g., a seventh face, the seventh surface **414** of FIG. **4**), an eighth surface **1126** (e.g., an eighth face), and a ninth surface **1128** (e.g., a ninth face). The eighth surface **1126** is contiguous with the seventh surface **1124** and the ninth surface **1128**. In the illustrated example of FIG. **11**, the seventh surface **1124** and the eighth surface **1126** are oblique relative to the axis **128**. The seventh surface **1124** is flush with (e.g., abuts and is positioned along a same geometric plane (e.g., the geometric plane **424** (FIG. **4**) as) the first surface **1112**. The ninth surface **1128** is substantially parallel to the axis **128**.

[0058] In the illustrated example of FIG. **11**, the eighth surface **1126** is in contact with the second surface **1114**, and the ninth surface **1128** is in contact with the third surface **1116**. The eighth surface **1126** applies a pre-load compression force to the second surface **1114** in a direction normal to the second surface **1114** (e.g., in a direction oblique relative to the axis **128**). Thus, the disc cover **1110** applies the pre-load compression force to a surface (e.g., the second surface **1114**) that is contiguous with a surface (e.g., the first surface **1112**) that contacts the seat **1102** to increase a contact area across which the pre-load is applied. As a result, the seat ring **1108** receives the pre-load across an increased distance between an inner and outer diameter thereof, which increases the pre-load force received and increases a pressure that the seat ring **1108** can withstand during operation. Moreover, as the second surface **1114** spans from the end of the first surface **1112** to the inner diameter of the seat ring **1104** such that the pre-load is applied in a same direction from the end of the first surface **1112** and the inner diameter of the seat ring **1104** to increase a uniformity with which the seat ring **1108** is pre-loaded.

[0059] FIG. **12** illustrates a portion of another example valve **1200** including a seat **1202** (e.g., the seat **114** (FIGS. **1-4**), the seat **814** (FIG. **8**)) and a valve plug **1204** in a closed position **1206**. The valve plug **1204** includes another example seat ring **1208**, and another example disc cover **1210**. The seat ring **1208** includes a first surface **1212** (e.g., a first face, a first bevel surface, the first surface **402** of FIG. **4**), a second surface **1214** (e.g., a second face, a second bevel surface, the second surface **404** of FIG. **4**), a third surface **1216** (e.g., a third face, the third surface **406** of FIG. **4**), a fourth surface **1218** (e.g., a fourth face), a fifth surface **1220** (e.g., a fifth face, a third bevel surface), and a sixth surface **1222** (e.g., a sixth face). In the illustrated example of FIG. **12**, the first surface **1212** is contiguous with the second surface **1214** and the sixth surface **1222**, and the second surface **1214** is contiguous with the third surface **1216**. In some examples, the fourth surface **1218**

is contiguous with the third surface **1216** and/or the fifth surface **1220**. In some examples, the fifth surface **1220** is contiguous with the sixth surface **1222**.

[0060] In the illustrated example of FIG. **12**, the first surface **1212**, the second surface **1214**, and the fifth surface **1220** are oblique relative to the axis **128** along which the valve plug **1204** translates. In some examples, the third surface **1216** is substantially perpendicular to the axis **128**. In some examples, the fourth surface **1218** and the sixth surface **1222** are substantially parallel to the axis **128**. In this example, the fifth surface **1220** is oriented at an angle greater than 90° relative to the sixth surface **1222**.

[0061] In the illustrated example of FIG. **12**, the first surface **1212** contacts the seat **1202** when the valve plug **1204** is in the closed position **1206**. Accordingly, fluid contacts and flows across the first surface **1212** when the valve plug **1204** is in an open position (e.g., the open position **200** of FIG. **2**). The second surface **1214**, the third surface **1216**, and the fourth surface **1218** are in contact with the disc cover **1210**. The fifth surface **1220** and the sixth surface **1222** are in contact with the disc holder **1206**.

[0062] In the illustrated example of FIG. **12**, the disc holder **1206** includes a seventh surface **1224** (e.g., a seventh face, a fourth bevel surface) and an eighth surface **1226** (e.g., an eighth face) contiguous with the seventh surface **1224**. The eighth surface **1226** is substantially parallel to the axis **128** and contacts the sixth surface **1222**. The seventh surface **1224** is oblique relative to the axis **128** and contacts the fifth surface **1220**. Specifically, the seventh surface **1224** is oriented at an angle greater than 90° relative to the eighth surface **1226**. As a result of the angular orientation of the fifth surface **1220** and the seventh surface **1224**, a size (e.g., a volume) of the seat ring **1208** is increased, which can enable the seat ring **1208** to withstand higher pressures. Additionally, the angular orientation of the fifth surface **1220** and the seventh surface **1224** increases a contact area therebetween. The increased contact area can improve sealing between the fifth surface **1220** and the seventh surface **1224** and/or improve a distribution of pressure in the seat ring **1208** that results from pre-loading to further prevent the seat ring **1208** from rupturing. In some examples, the fifth surface **1220** includes raised portions (e.g., the raised portions **434** (FIG. **4**)), and the seventh surface **1224** includes the protrusions (e.g., the protrusions **436** (FIG. **4**)) that mesh with the raised portions. In such examples, the raised portions and the protrusions extend along a direction normal to the seventh surface **1224** and the fifth surface **1220**. In some examples, the valve plug **1204** includes an O-ring (e.g., the O-ring **818** (FIG. **8**)) positioned between the fifth surface **1220** and the seventh surface **1224**.

[0063] In the illustrated example of FIG. **12**, the disc cover **1210** includes a ninth surface **1228** (e.g., a ninth face, a fifth bevel surface, the seventh surface **414** of FIG. **4**), a tenth surface **1230** (e.g., a tenth face, a sixth bevel surface, the eighth surface **416** of FIG. **4**), an eleventh surface **1232** (e.g., an eleventh face, the ninth surface **418** of FIG. **4**), and a twelfth surface **1234** (e.g., a twelfth face). The tenth surface **1230** is contiguous with the ninth surface **1228** and the eleventh surface **1232**, and the eleventh surface **1232** is contiguous with the twelfth surface **1234**. The ninth surface **1228** and the tenth surface **1230** are oblique relative to the axis **128**. The eleventh surface **1232** is substantially perpendicular to the axis **128**. The twelfth surface **1234** is substantially parallel to the axis **128**.

[0064] In the illustrated example of FIG. **12**, the ninth surface **1228** is flush with (e.g., abuts and is positioned along a same geometric plane (e.g., the geometric plane **424** (FIG. **4**) as) the first surface **1212**. The tenth surface **1230** is in contact with the second surface **1214**; the eleventh surface **1232** is in contact with the third surface **1216**; and the twelfth surface is in contact with the fourth surface **1218**. The tenth surface **1230** applies a first pre-load force to the second surface **1214**, and the eleventh surface **1232** applies a second pre-load force to the third surface **1216** to improve seal that the seat ring **1208** provides in the closed position **1206** and maintain a structural integrity thereof against high pressures, as discussed in further detail above.

[0065] “Including” and “comprising” (and all forms and tenses thereof) are used herein to be open

ended terms. Thus, whenever a claim employs any form of “include” or “comprise” (e.g., comprises, includes, comprising, including, having, etc.) as a preamble or within a claim recitation of any kind, it is to be understood that additional elements, terms, etc., may be present without falling outside the scope of the corresponding claim or recitation. As used herein, when the phrase “at least” is used as the transition term in, for example, a preamble of a claim, it is open-ended in the same manner as the term “comprising” and “including” are open ended. The term “and/or” when used, for example, in a form such as A, B, and/or C refers to any combination or subset of A, B, C such as (1) A alone, (2) B alone, (3) C alone, (4) A with B, (5) A with C, (6) B with C, or (7) A with B and with C. As used herein in the context of describing structures, components, items, objects and/or things, the phrase “at least one of A and B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B.

Similarly, as used herein in the context of describing structures, components, items, objects and/or things, the phrase “at least one of A or B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. As used herein in the context of describing the performance or execution of processes, instructions, actions, activities, etc., the phrase “at least one of A and B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. Similarly, as used herein in the context of describing the performance or execution of processes, instructions, actions, activities, etc., the phrase “at least one of A or B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B.

[0066] As used herein, singular references (e.g., “a”, “an”, “first”, “second”, etc.) do not exclude a plurality. The term “a” or “an” object, as used herein, refers to one or more of that object. The terms “a” (or “an”), “one or more”, and “at least one” are used interchangeably herein.

Furthermore, although individually listed, a plurality of means, elements, or actions may be implemented by, e.g., the same entity or object. Additionally, although individual features may be included in different examples or claims, these may possibly be combined, and the inclusion in different examples or claims does not imply that a combination of features is not feasible and/or advantageous.

[0067] As used herein, unless otherwise stated, the term “above” describes the relationship of two parts relative to Earth. A first part is above a second part, if the second part has at least one part between Earth and the first part. Likewise, as used herein, a first part is “below” a second part when the first part is closer to the Earth than the second part. As noted above, a first part can be above or below a second part with one or more of: other parts therebetween, without other parts therebetween, with the first and second parts touching, or without the first and second parts being in direct contact with one another.

[0068] As used in this patent, stating that any part (e.g., a layer, film, area, region, or plate) is in any way on (e.g., positioned on, located on, disposed on, or formed on, etc.) another part, indicates that the referenced part is either in contact with the other part, or that the referenced part is above the other part with one or more intermediate part(s) located therebetween.

[0069] As used herein, connection references (e.g., attached, coupled, connected, and joined) may include intermediate members between the elements referenced by the connection reference and/or relative movement between those elements unless otherwise indicated. As such, connection references do not necessarily infer that two elements are directly connected and/or in fixed relation to each other. As used herein, stating that any part is in “contact” with another part is defined to mean that there is no intermediate part between the two parts.

[0070] Unless specifically stated otherwise, descriptors such as “first,” “second,” “third,” etc., are used herein without imputing or otherwise indicating any meaning of priority, physical order, arrangement in a list, and/or ordering in any way, but are merely used as labels and/or arbitrary names to distinguish elements for ease of understanding the disclosed examples. In some examples, the descriptor “first” may be used to refer to an element in the detailed description, while the same

element may be referred to in a claim with a different descriptor such as “second” or “third.” In such instances, it should be understood that such descriptors are used merely for identifying those elements distinctly within the context of the discussion (e.g., within a claim) in which the elements might, for example, otherwise share a same name.

[0071] As used herein, “approximately” and “about” modify their subjects/values to recognize the potential presence of variations that occur in real world applications. For example, “approximately” and “about” may modify dimensions that may not be exact due to manufacturing tolerances and/or other real world imperfections as will be understood by persons of ordinary skill in the art. For example, “approximately” and “about” may indicate such dimensions may be within a tolerance range of $\pm 10\%$ unless otherwise specified herein.

[0072] As used herein in the context of describing the position and/or orientation of a first object (e.g., a first surface, a first plane, a first axis, etc.) relative to a second object (e.g., a second surface, a second plane, a second axis, etc.) the term “substantially perpendicular” encompasses the term perpendicular and more broadly encompasses a meaning whereby the first object is positioned and/or oriented relative to the second object at an absolute angle of no more than five degrees (5° from perpendicular. For example, a first axis that is substantially perpendicular to a second axis is positioned and/or oriented relative to the second axis at an absolute angle of no more than five degrees (5° from perpendicular.

[0073] As used herein in the context of describing the position and/or orientation of a first object (e.g., a first surface, a first plane, a first axis, etc.) relative to a second object (e.g., a second surface, a second plane, a second axis, etc.) the term “substantially parallel” encompasses the term parallel and more broadly encompasses a meaning whereby the first object is positioned and/or oriented relative to the second object at an absolute angle of no more than five degrees (5° from parallel. For example, a first axis that is substantially parallel to a second axis is positioned and/or oriented relative to the second axis at an absolute angle of no more than five degrees (5° from parallel.

[0074] From the foregoing, it will be appreciated that example systems, apparatus, articles of manufacture, and methods have been disclosed that increase a pressure at which a valve plug maintains a sealing capability while maintaining a flush flow path boundary to avoid undesirable flow characteristics in a valve.

[0075] Example valve sealing structures are disclosed herein. Further examples and combinations thereof include the following:

[0076] Example 1 includes an apparatus comprising a valve body including a seat, and a valve plug including a disc holder including a recess, a seat ring positioned in the recess, the seat ring including a first surface and a second surface contiguous with the first surface, the first surface to contact the seat when the valve plug is in a closed position, and a disc cover coupled to the disc holder, the disc cover including a third surface and a fourth surface contiguous with the third surface, the third surface to be flush with the first surface, the fourth surface to contact the second surface and apply an axial pre-load force to the seat ring. [0077] Example 2 includes the apparatus of example 1, wherein the axial pre-load force is a first axial pre-load force in a first direction, wherein the seat ring includes a fifth surface, and wherein the disc cover includes a sixth surface, wherein the sixth surface contacts the fifth surface to apply a second axial pre-load force to the seat ring in a second direction different than the first direction. [0078] Example 3 includes the apparatus of example 2, wherein the fifth surface is contiguous with the second surface, and wherein the sixth surface is contiguous with the fourth surface. [0079] Example 4 includes the apparatus of example 1, wherein the seat ring includes a fifth surface in contact with the disc holder, and wherein the fifth surface includes raised portions. [0080] Example 5 includes the apparatus of example 4, wherein the raised portions are first serrations, and wherein the disc holder includes second serrations that mesh with the first serrations. [0081] Example 6 includes the apparatus of example 4, wherein the raised portions are concentric serrations. [0082] Example 7 includes the apparatus of example 1, wherein the second surface includes a bevel that is nonparallel to a longitudinal axis of the valve plug. [0083] Example 8 includes the apparatus of

example 1, wherein the second surface includes a bevel that is oblique to a longitudinal axis of the valve plug. [0084] Example 9 includes an apparatus comprising a valve body including an inner wall that defines a seat, the seat positioned around an orifice, a disc holder, a disc cover coupled to the disc holder, the disc cover including a first surface and a second surface that is contiguous with the first surface, the first surface at least partially facing the inner wall, and a seat ring positioned between the disc cover and the disc holder, the seat ring including a third surface and a fourth surface contiguous with the third surface, the third surface to contact the second surface, the third surface including an angular orientation that is non-perpendicular to a plane defined by the orifice, the fourth surface flush with the first surface. [0085] Example 10 includes the apparatus of example 9, wherein the seat ring, the disc holder, and the disc cover are movable between a first position and a second position, the fourth surface to contact the seat in the first position to block fluid from flowing between the seat and the seat ring, the fourth surface separated from the seat in the second position. [0086] Example 11 includes the apparatus of example 9, wherein at least a portion of the second surface faces the disc holder. [0087] Example 12 includes the apparatus of example 9, wherein the angular orientation of the third surface is nonparallel to the plane defined by the orifice. [0088] Example 13 includes the apparatus of example 9, wherein the seat ring includes a fifth surface, wherein the disc cover includes a sixth surface that contacts the fifth surface, wherein the disc cover compresses the seat ring with a first force on the fifth surface and a second force on the third surface. [0089] Example 14 includes the apparatus of example 13, wherein the fifth surface and the sixth surface are substantially parallel to the plane defined by the orifice. [0090] Example 15 includes the apparatus of example 14, wherein the fifth surface is contiguous with the third surface, and wherein the sixth surface is contiguous with the second surface. [0091] Example 16 includes the apparatus of example 13, wherein the first force is approximately normal to the third surface and the second force is approximately normal to the fifth surface. [0092] Example 17 includes the apparatus of example 13, wherein the disc cover includes threads to couple the disc cover to the disc holder, and wherein the threads enable conversion of a rotation of the disc cover to linear movement that applies the first force and the second force. [0093] Example 18 includes the apparatus of example 9, wherein the seat ring includes a fifth surface in contact with the disc holder, and wherein the fifth surface includes an indent. [0094] Example 19 includes the apparatus of example 18, wherein the disc holder includes a protrusion that extends into a cavity defined by the indent. [0095] Example 20 includes an apparatus comprising a seat ring including a first face, a second face, and a third face, the first face and the third face contiguous with the second face, the first face to contact a disc holder, the third face to contact a disc cover, the second face flush with a surface of the disc cover, the seat ring movable along an axis between a first position and a second position, the second face to contact a perimeter of an orifice when the seat ring is in the first position, the second face including a first bevel that is nonparallel to the axis, the third face including a second bevel that is nonparallel to the axis.

[0096] The following claims are hereby incorporated into this Detailed Description by this reference. Although certain example systems, apparatus, articles of manufacture, and methods have been disclosed herein, the scope of coverage of this patent is not limited thereto. On the contrary, this patent covers all systems, apparatus, articles of manufacture, and methods fairly falling within the scope of the claims of this patent.

Claims

1. An apparatus comprising: a valve body including a seat; and a valve plug including: a disc holder including a recess; a seat ring positioned in the recess, the seat ring including a first surface and a second surface contiguous with the first surface, the first surface to contact the seat when the valve plug is in a closed position; and a disc cover coupled to the disc holder, the disc cover including a third surface and a fourth surface contiguous with the third surface, the third surface to

be flush with the first surface, the fourth surface to contact the second surface and apply an axial pre-load force to the seat ring.

2. The apparatus of claim 1, wherein the axial pre-load force is a first axial pre-load force in a first direction, wherein the seat ring includes a fifth surface, and wherein the disc cover includes a sixth surface, wherein the sixth surface contacts the fifth surface to apply a second axial pre-load force to the seat ring in a second direction different than the first direction.

3. The apparatus of claim 2, wherein the fifth surface is contiguous with the second surface, and wherein the sixth surface is contiguous with the fourth surface.

4. The apparatus of claim 1, wherein the seat ring includes a fifth surface in contact with the disc holder, and wherein the fifth surface includes raised portions.

5. The apparatus of claim 4, wherein the raised portions are first serrations, and wherein the disc holder includes second serrations that mesh with the first serrations.

6. The apparatus of claim 4, wherein the raised portions are concentric serrations.

7. The apparatus of claim 1, wherein the second surface includes a bevel that is nonparallel to a longitudinal axis of the valve plug.

8. The apparatus of claim 1, wherein the second surface includes a bevel that is oblique to a longitudinal axis of the valve plug.

9. An apparatus comprising: a valve body including an inner wall that defines a seat, the seat positioned around an orifice; a disc holder; a disc cover coupled to the disc holder, the disc cover including a first surface and a second surface that is contiguous with the first surface, the first surface at least partially facing the inner wall; and a seat ring positioned between the disc cover and the disc holder, the seat ring including a third surface and a fourth surface contiguous with the third surface, the third surface to contact the second surface, the third surface including an angular orientation that is non-perpendicular to a plane defined by the orifice, the fourth surface flush with the first surface.

10. The apparatus of claim 9, wherein the seat ring, the disc holder, and the disc cover are movable between a first position and a second position, the fourth surface to contact the seat in the first position to block fluid from flowing between the seat and the seat ring, the fourth surface separated from the seat in the second position.

11. The apparatus of claim 9, wherein at least a portion of the second surface faces the disc holder.

12. The apparatus of claim 9, wherein the angular orientation of the third surface is nonparallel to the plane defined by the orifice.

13. The apparatus of claim 9, wherein the seat ring includes a fifth surface, wherein the disc cover includes a sixth surface that contacts the fifth surface, wherein the disc cover compresses the seat ring with a first force on the fifth surface and a second force on the third surface.

14. The apparatus of claim 13, wherein the fifth surface and the sixth surface are substantially parallel to the plane defined by the orifice.

15. The apparatus of claim 14, wherein the fifth surface is contiguous with the third surface, and wherein the sixth surface is contiguous with the second surface.

16. The apparatus of claim 13, wherein the first force is approximately normal to the third surface and the second force is approximately normal to the fifth surface.

17. The apparatus of claim 13, wherein the disc cover includes threads to couple the disc cover to the disc holder, and wherein the threads enable conversion of a rotation of the disc cover to linear movement that applies the first force and the second force.

18. The apparatus of claim 9, wherein the seat ring includes a fifth surface in contact with the disc holder, and wherein the fifth surface includes an indent.

19. The apparatus of claim 18, wherein the disc holder includes a protrusion that extends into a cavity defined by the indent.

20. An apparatus comprising: a seat ring including a first face, a second face, and a third face, the first face and the third face contiguous with the second face, the first face to contact a disc holder,

the third face to contact a disc cover, the second face flush with a surface of the disc cover, the seat ring movable along an axis between a first position and a second position, the second face to contact a perimeter of an orifice when the seat ring is in the first position, the second face including a first bevel that is nonparallel to the axis, the third face including a second bevel that is nonparallel to the axis.
